NARAYANI

CYBERSECURITY

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Portfolio • LinkedIn • GitHub • Google Scholar

PROFILE

M.Tech Cybersecurity researcher at IGDTUW, working under the MeitY–ISEA Phase-III project. Focused on digital forensics, DFIR, anti-forensics detection, threat analysis and evidence integrity. Experienced in developing Python-based forensic and security proof-of-concept solutions. Research interests include AI-assisted digital investigation, privacy-preserving image forgery detection and Federated Learning. EC-Council Certified Hacking Forensic Investigator (CHFI).

TECHNICAL EXPERTISE

Security Operations	Threat Hunting, Threat Analysis, MITRE ATT&CK, Incident Response, Network Reconnaissance, Vulnerability Assessment
Digital Forensics	DFIR, Disk Forensics, Memory Forensics, Metadata Analysis, Anti-Forensics, Artifact Analysis, Timeline Reconstruction, Chain of Custody
Security Tools	Nmap, Wireshark, Metasploit, Kali Linux, Burp Suite, Autopsy, FTK, Volatility
Programming	Python, Bash, C, C++, Java, SQL
Development	Flask, FastAPI, REST APIs, SQLite, MySQL, Docker, Git, GitHub, Linux, Windows
AI & Research	Machine Learning, Deep Learning, CNN, ResNet50, Computer Vision, Federated Learning, FedAvg, FedProx, FedYogi

PROFESSIONAL EXPERIENCE

Research Scholar — **Aug 2025 – Present**
MeitY–ISEA Project Phase-III, IGDTUW — New Delhi

- Contribute to cybersecurity and digital forensics research, including the design, development and documentation of security-focused proof-of-concept solutions.
- Work on digital evidence analysis, anti-forensics detection, threat analysis and AI-assisted investigation workflows; support technical workshops and mentoring.

Makerspace Facilitator & ERP Coordinator — **Jul 2024 – May 2025**
Heritage Xperiential Learning School — Noida

- Delivered practical technology instruction to 1,000+ students, guided 50+ student projects and coordinated ERP updates and academic result publication.

PROJECTS

Digital Wipe Detector Framework — **View Repository**

Developed a forensic framework to investigate secure-deletion indicators, evidence wiping and potential digital artifact tampering.

Deepfake–AFDF Framework — **View Repository**

Designed a research framework for examining digital evidence manipulation, metadata alteration, artifact obfuscation and anti-forensic behaviour.

Federated Image Forgery Detection — **View Repository**

Explored privacy-preserving image forgery detection using Federated Learning, FedYogi, CNN and ResNet50.

Secure Student Management System — **View Repository**

Built a secure Flask application with password hashing, RBAC, email OTP, MFA, session management, audit logging and role-based Admin, Faculty and Student workflows.

EDUCATION

M.Tech – Information Technology (Cybersecurity) — **2025 – 2027**
Indira Gandhi Delhi Technical University for Women — CGPA: 9.0/10

B.Tech – Electronics and Communication Engineering — **2021 – 2024**
Ganga Technical Campus, MDU, Rohtak — 75%

Diploma – Electronics and Communication Engineering — **2017 – 2020**
Pusa Institute of Technology, BTE Delhi — 75%

CERTIFICATIONS & RECOGNITION

- **CHFI – EC-Council:** Certified Hacking Forensic Investigator, February 2026.
- **NFSU Hackathon:** 3rd Place; INR 50,000 prize for an Anti-Forensic Evasion Framework project.
- **HERizon Hackathon:** 3rd Place, Defense Theme.
- **SMARTCOM 2026:** Federated Learning research paper accepted.
- **ISEA–ISAP 2026, IIT Madras:** Selected for academic poster presentation.